

External Human-Machine Interface for Robotaxis

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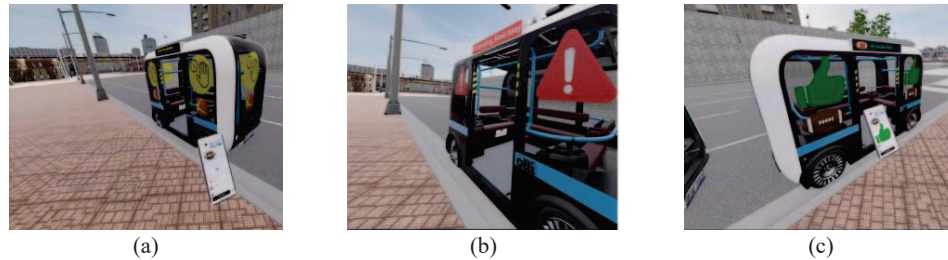


Figure 1: The proposed external human-machine interface (eHMI) uses exterior of the vehicle to (a) inform the passengers to wait until it's safe to approach the robotaxi (b) or warn the passengers if the distance between them and the robotaxi is dangerous (c) or inform them when the robotaxi is ready for boarding.

ABSTRACT

The increasing popularity of automated vehicles and shared mobility services has led to the development of robotaxis as a new mode of transportation. To ensure a positive and safe user experience, effective human-machine interfaces (HMI) are needed. This paper focuses on the design of external HMIs (eHMIs) for robotaxis, with a focus on using multiple modalities to enhance the user experience and improve communication between passengers and the automated system. By reviewing the current state of the art and key design considerations for eHMIs, this paper aims to contribute to the development of effective and user-friendly eHMIs for robotaxis with respect to rider-vehicle match process, passenger safety and security. The research team prototyped eHMIs in an immersive virtual environment using a virtual reality headset. The pilot study with five participants suggests a preference for the eHMI in robotaxi pickups, and its potential to enhance user experience and safety.

Keywords: Automated Vehicles; eHMIs; Robotaxis, Human-computer interaction, User interface application. **Index Terms:** H.5 [Information Interfaces and Presentation]: H.5.1: Multimedia Information Systems — Artificial, Augmented, and Virtual Realities.

1 INTRODUCTION

Current HMIs for shared mobility services primarily rely on mobile applications, touchscreens, and other visual displays to provide

information to passengers and receive input. While these interfaces have proven to be effective in providing basic information, they may not be suitable for all users and scenarios. Moreover, the passenger-vehicle picking process, where a passenger identifies and boards the robotaxi, is an important aspect of the overall user experience. How does the passenger know it's safe to approach the vehicle and how does the passenger know which robotaxi to board? There have been several concerns for passenger safety during pickup with ride-hailing services such as Uber and Lyft. Currently the passengers have a human driver to interact with before being picked up. The absence of a vehicle driver in robotaxis makes the HMI plays a crucial role in this process by providing information about the vehicle to ensure a safe picking process. For robotaxis to successfully communicate with passengers who are about to be picked up, an external HMI may be the most effective. It is known that people understand the intentions of AVs better when those intentions are communicated via external HMIs [1]. Additionally, research has shown that the use of eHMIs on AVs may increase efficiency and safety of interaction with road users such as pedestrians [2], which opens the door to future research in eHMI development and interaction potential. It is also known that users appreciate content which helps to provide contextual information for dealing with the AV interaction [3]. However, to date, there has not been much research work done around how eHMIs can influence the process of picking up passengers in shared automated vehicles.

2 OBJECTIVE

The research team aimed to contribute to the development of effective and user-friendly eHMIs for robotaxis. For this study, we propose eHMIs to support the passenger-vehicle match & boarding process and evaluate usability of the proposed user interface as compared to a traditional mobile-phone-based user interface.

3 METHOD

To achieve our objectives, we relied on the user experience lifecycle framework proposed by Hartson and Pyla (2012) [4], which consists of four phases: requirements analysis, design, prototyping, and evaluation. To better understand the needs of the

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users, we conducted interviews and reviewed the literature for any related design concepts or guidelines that could inform our design. A structured interview was conducted with users of ride-hailing services to learn more about the current passenger-vehicle match and boarding process.

The findings of the requirements analysis phase resulted in an eHMI design (as shown in Figure 1) that utilizes both visual and Audio components. Visual components were used to help users identify the vehicle, inform the passenger about the status of the vehicle (operating, parked), while in safety critical situations both visuals and audio were used to warn the users if they are close to the vehicle while it's still operating. Audio components were only used in high-risk situations. The design also takes into consideration that no private information should be displayed, in response to the security concerns that were raised in the interviews.

Using the Unreal Engine, we developed a virtual environment to simulate real-world scenarios. This virtual setting, modeled after an urban landscape with roads, traffic, and sidewalks where participants were placed. To provide participants with a highly immersive experience, we relied on the Varjo XR-3 VR headset [5], capable of hand tracking. The robotaxi eHMI design was implemented in the prototype, along with a phone UI that allowed participants to open the app, order the ride, and read information about their ride. An invisible collision box called the "dangerous zone" was created around the robotaxi for evaluation, along with other automatic data collection methods. The dangerous zone was only activated if it was not safe to get close to the vehicle while it was still operating.

To evaluate our design, a within-subject pilot study was conducted with five participants, where the independent variable was the use of eHMI where all participants experienced two conditions (eHMI + mobile phone vs. mobile phone only). During the study participants were asked to order a ride using the mobile phone, wait for their ride to arrive, identify their ride, and board the ride by physically walking and enter the vehicle when it's safe. The concepts of interest were safety, usability, and performance. These concepts were operationalized using different measures. Safety was measured based on two indicators: (1) the number of times participants entered the dangerous zone (2) time spent inside the dangerous zone. Performance was measured based on (1) boarding time: time spent boarding from the moment the UI indicated it's safe to board and (2) error rate: the number of times users boarded the wrong vehicle. Usability was measured using the System Usability Scale (SUS) questionnaire [6].

4 RESULTS

The results indicated a significant difference in boarding time when using external HMI as compared to a traditional user interface presented on a smart phone. The mean boarding time for eHMI was 14.06 seconds, compared to 22.37 seconds for the mobile phone. Additionally, there was a significant difference in the time spent in the danger zone. The mean time spent in the danger zone without eHMI was 21.82 seconds, while with eHMI, it was 3.9 seconds. These significant time differences may be attributed to the user-friendly layout of the external HMI and the avatars utilized by individuals. The results of the SUS questionnaire indicate that users preferred eHMI over the traditional mobile-phone-based interface.

5 CONCLUSION AND LIMITATIONS

The pilot study suggests a preference for eHMI in robotaxi pickups, and there is potential for eHMI to enhance safety and improve the user experience. Limitations include sample size, space constraints and the VR headset's restricted cable length. Despite these limitations, the pilot study suggests eHMI's potential to improve

safety and user experience in robotaxi pickups, paving the way for future implementation and increased accessibility.

6 FUTURE WORK

We plan to conduct future studies in more spacious environments, allowing users to move freely and interact with the simulated scenario without constraints. Additionally, the small sample size is acknowledged as a limitation, and we plan to conduct a full-scale study with a larger and more diverse participant pool to ensure a more robust analysis of the impact of eHMI in robotaxi pickups. By addressing these limitations, we aim to refine the experimental conditions and gather more comprehensive data, thereby enhancing the validity and reliability of our findings in future research endeavors.

In subsequent studies, we would like to evaluate the impact of visual and audio components of eHMIs on the safety not only on passengers but also on other road users like drivers, cyclists, and pedestrians. By understanding the effects of eHMI designs on all road users, we will be able to create more universally effective solutions. This comprehensive approach will not only contribute to a safer environment for various types of road users but also promote the widespread adoption of autonomous vehicles in the future.

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